Line integrals for Vector field - Green's theorem

1) Line integral for a vector field

Defn: Let $\vec{F} : \mathbb{R}^n \to \mathbb{R}^n$ be a vector field and $\vec{r} : [a;b] \to \mathbb{R}^n$ a vector function. The line integral of $\vec{F}$ over $\mathcal{C} = \text{Range}(\vec{r})$ is

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$$

scalar product

Proposition:

$\vec{r}'(t) = \frac{d\vec{r}}{dt}$

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)\, dt$$

scalar product

and

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} \vec{F} \cdot \vec{T}\, ds$$

scalar product

and

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}} P(x,y)\, dx + Q(x,y)\, dy$$

if $n = 2$ and $\vec{F} = \langle P, Q \rangle$

Remark: • $\vec{T} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$

$$\vec{F} \cdot \vec{T}\, ds = \vec{F}(\vec{r}(t)) \cdot \frac{\vec{r}'(t)}{|\vec{r}'(t)|} \, |\vec{r}'(t)|\, dt$$

- $\vec{r}(t) = \langle x(t), y(t) \rangle$

$\frac{d\vec{r}}{dt} = \langle \frac{dx}{dt}, \frac{dy}{dt} \rangle$

$\vec{F} \cdot d\vec{r} = \langle P, Q \rangle \cdot \langle dx, dy \rangle$

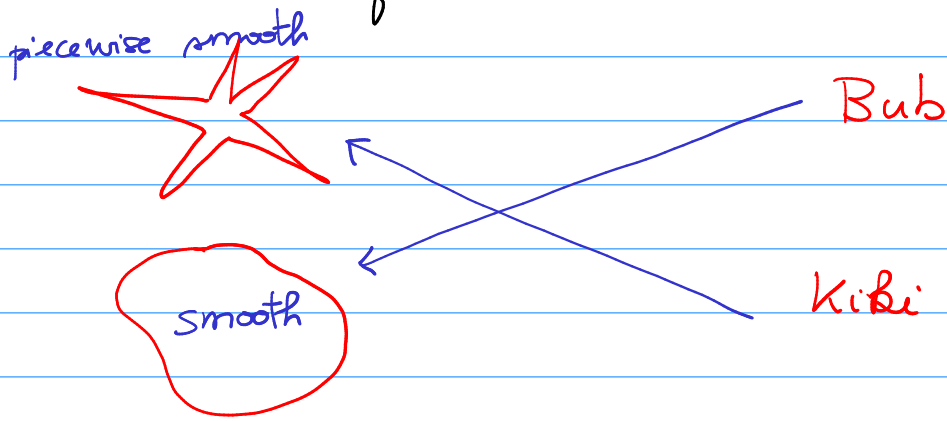
$= P\,dx + Q\,dy$

2) Fundamental theorem

Curves :

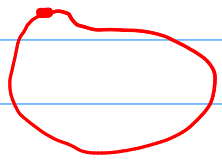
Defn : A curve $\mathcal{C} = \text{Range}(\vec{r})$ is

- smooth if $\vec{r}\,'(t)$ exists and is continuous.



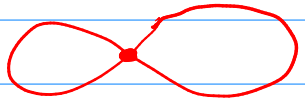
$\vec{r} : [a;b] \to \mathbb{R}^n$

- closed if $\vec{r}(a) = \vec{r}(b)$



- simple if it doesn't cross itself:

If there are no t_1, t_2 with $a < t_1 < t_2 < b$ such that $\vec{r}(t_1) = \vec{r}(t_2)$



not simple

Theorem : If $\vec{F}$ is such that $\vec{F} = \nabla f$ and $\mathcal{C}$ is piecewise smooth, then

$$\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

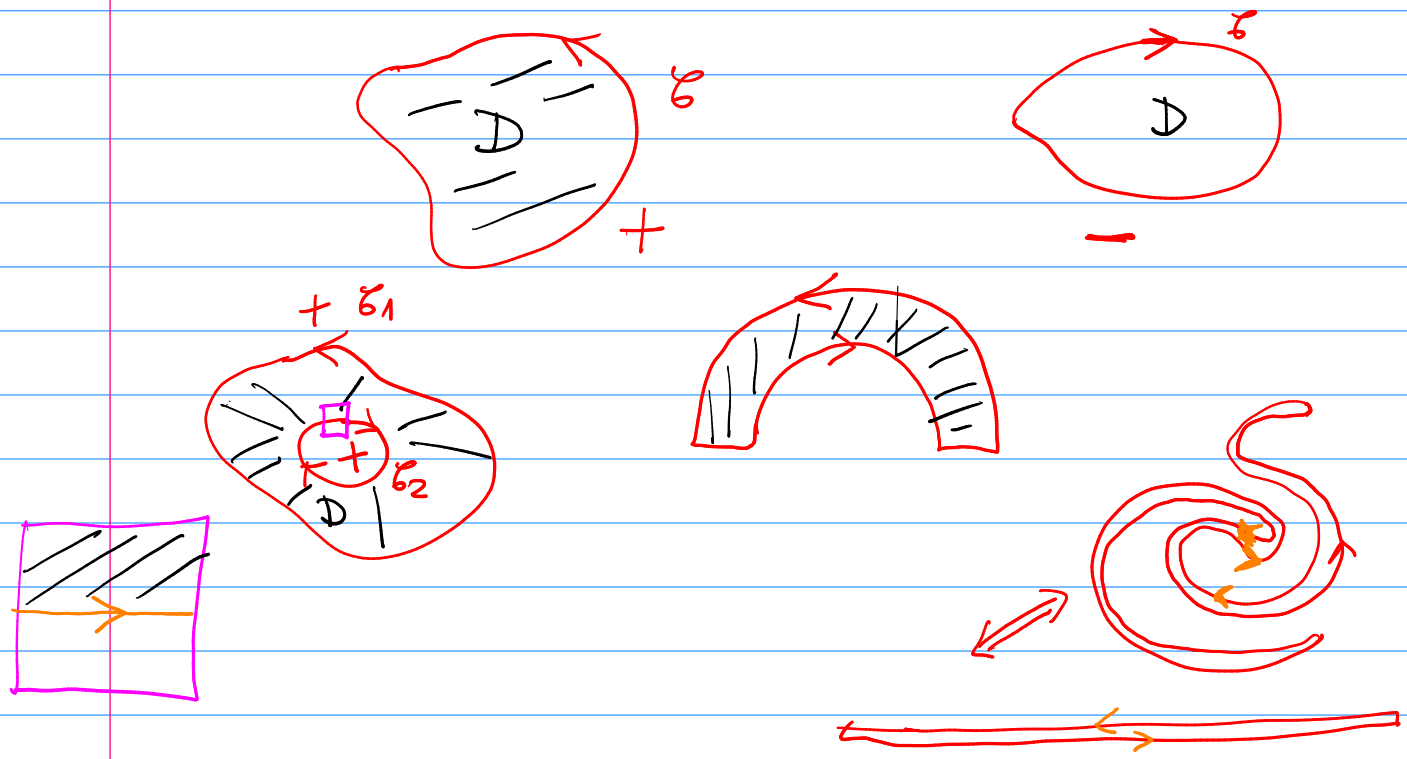
where $\vec{r} : [a;b] \to \mathbb{R}^n$ and $\mathcal{C} = \text{Range}(\vec{r})$

Remark: If $\mathcal{C}$ is closed, then $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = 0$

Defn: If $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r} = 0$ for all closed curves in a region D, then we say that $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$ is independent of path.

3) Green's Theorem

a) **Orientation:** A region D which is delimited by a curve $\mathcal{C} = \text{Range}(\vec{r})$ is positively oriented if $\vec{r}(t)$ runs $\mathcal{C}$ in counterclockwise manner, with regard to D (D at the left of the curve)

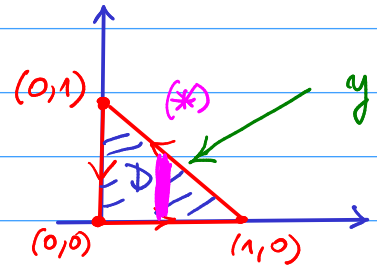


Theorem: $\mathcal{C}$ is positively oriented, piecewise smooth, simple and closed. If $\vec{F} = \langle P, Q \rangle$ and D is the region delimited by $\mathcal{C}$, then

$$\int_{\mathcal{C}} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

b) Examples

(i) $\int_{\mathcal{C}} \frac{\sqrt{x}}{1+x^2}\,dx + xy\,dy$ where $\mathcal{C}$ is the triangle with vertices: $(0,0)$, $(1,0)$ and $(0,1)$



$$\vec{F}(x,y) = \left\langle \underbrace{\frac{\sqrt{x}}{1+x^2}}_{P}, \underbrace{xy}_{Q} \right\rangle$$

$$GT \Rightarrow \int_{\mathcal{C}} \frac{\sqrt{x}}{1+x^2}\,dx + xy\,dy = \iint_D \underbrace{(y - 0)}_{(*)}\,dA$$

$$\Rightarrow D = \{(x,y) \mid 0 \le x \le 1 \text{ and } 0 \le y \le 1-x\}$$

$$\iint_D y\,dA = \int_0^1 \left(\int_0^{1-x} y\,dy\right) dx.$$

($\vec{F} = \langle P, Q \rangle$ and $\mathcal{C}$ given)

(ii) Area of a geometric shape

(D is given $\Rightarrow$ $\vec{F} = \langle P, Q \rangle$ has to be identified / created)



$A :=$ area of D

$$A = \iint_D 1\,dA = \int_{\mathcal{C}} P\,dx + Q\,dy$$

$$1 = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Choice of P and Q:

P	x	$-y$	$-\frac{1}{2}y$
Q	x	y	$\frac{1}{2}x$

(with $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$)

$\partial P/\partial y$	0	-1	$-\frac{1}{2}$
$\partial Q/\partial x$	1	0	$\frac{1}{2}$

Important

- Find the area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. $\quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- $A = \iint_D 1 \cdot dA$
- Choose $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$
  $\Rightarrow \vec{F} = \langle -\frac{1}{2}y, \frac{1}{2}x \rangle$
- C parametrised by:
  $\begin{cases} x = 3\cos(t) \\ y = 2\sin(t) \end{cases} \quad t = [0; 2\pi]$

$$\Rightarrow A = \int_C P\,dx + Q\,dy = \int_0^{2\pi} \left(3\sin^2(t) + 3\cos^2(t) \right) dt$$
$$= 6\pi \quad (\pi ab)$$
$$\| \quad 3 \times 2$$

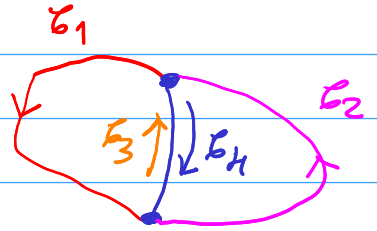
with $\begin{cases} dx = -3\sin(t)\,dt \\ dy = 2\cos(t)\,dt \end{cases}$

and

$\begin{cases} P \cdot dx = \left(-\frac{1}{2}(2\sin(t))\right)(-3\sin(t))\,dt \\ Q\,dy = \frac{1}{2}(3\cos(t))(2\cos(t))\,dt \end{cases}$

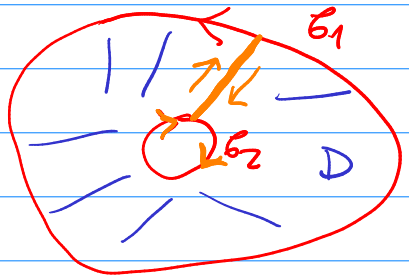
c) Combination of curves

$\vec{F} = \langle P, Q \rangle$ and $\mathcal{C}$ given.



$$\int_{\mathcal{C}_3} \vec{F} \cdot d\vec{r} = -\int_{\mathcal{C}_4} \vec{F} \cdot d\vec{r}$$

$$\int_{\mathcal{C}_1} \vec{F}\, d\vec{r} + \int_{\mathcal{C}_2} \vec{F}\, d\vec{r} + \int_{\mathcal{C}_3} \vec{F} \cdot d\vec{r} + \int_{\mathcal{C}_4} \vec{F} \cdot d\vec{r} = \int_{\mathcal{C}_1} \vec{F} \cdot d\vec{r} + \int_{\mathcal{C}_2} \vec{F} \cdot d\vec{r}$$



$$\int_{\mathcal{C}_1} \vec{F} \cdot d\vec{r} + \int_{\mathcal{C}_2} \vec{F} \cdot d\vec{r}$$